

Article

Vortex-Induced Vibration Analysis of FRP Composite Risers Using Multivariate Nonlinear Regression

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Abstract

Marine risers are essential for offshore resource extraction, yet traditional metal risers encounter limitations in deep-sea applications due to their substantial weight. Fiber-reinforced polymer (FRP) composites offer a promising alternative with advantages including low density and enhanced corrosion/fatigue resistance. However, FRP risers remain susceptible to fatigue damage from vortex-induced vibration (VIV). Therefore, this study investigated VIV behavior of FRP composite risers considering the coupled effect of tensile-flexural moduli, top tensions, slenderness ratios, and flow velocities. Through an orthogonal experimental design, eighteen cases were analyzed using multivariate nonlinear fitting. Results indicated that FRP composite risers exhibited larger vibration amplitudes than metal counterparts, with amplitudes increasing to both riser length and flow velocity. It was also found that the optimized FRP configuration demonstrated enhanced fiber strength utilization. Parameter coupling analysis revealed that the multivariate nonlinear fitting model achieved sufficient accuracy when incorporating two coupled parameters, with the most significant interaction occurring between flexural modulus and top tension.

Keywords: marine riser; FRP composite riser; vortex-induced vibration; multivariate nonlinear fitting; coupling effect



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1. Introduction

The rapid development of the global economy has led to an increasing demand for energy, and more and more countries are progressively shifting their focus from land-based to offshore oil and gas extraction, particularly in deep-sea environments. The marine riser serves as a crucial link between offshore platforms and subsea oil and gas production systems [1]. With the development of oil and gas exploration and extraction in deep-sea environments, traditional metal risers have experienced a significant increase in self-weight. This inherent limitation has hindered the industry's progress. In contrast, fiber-reinforced polymer (FRP) composite marine risers possess a lower density and superior mechanical properties, allowing for a reduction in self-weight, which subsequently decreases the top

tension requirements and operating costs of existing offshore platforms. Additionally, FRP composite marine risers exhibit excellent corrosion resistance, high vibration damping, and fatigue resistance [2,3], which contribute to lower maintenance and operational costs. Particularly in the last decade, the application of composites in marine engineering has expanded significantly, leading to more attempts aimed at developing FRP composite risers. With the rapid advancement of computer technology, the integrated methods of Computational Fluid Dynamics (CFD) and fluid–structure interaction (FSI) have gradually emerged [4].

As the more promising choice, comparative research between the FRP composite riser and the traditional metal riser has become one of the research hotspots. Tan et al. conducted a comprehensive analysis on both local and global scales to investigate the fluid–structure coupling response of both types of risers at a depth of 1500 m, which presented the greater margin of safety of the composite risers. For the performance of the liner for the composite riser [5], the titanium liner was proven to be a better choice than steel and aluminum liners. The power response models of metal and FRP composite risers were studied under various combinations of operational parameters [6]. Moreover, the distinct vortex-induced vibration (VIV) characteristics under identical flow conditions of metal and FRP composite risers were investigated [7], and the results showed that FRP composite risers led to greater deformation and a reduced stress response compared to traditional metal risers. The VIV performance, i.e., the intrinsic frequency, overall displacement, and overall stress of the FRP composite riser and the X80 steel riser, was obtained using CFD and FSI methods [8], and the impacts of various parameters were studied using Grey Theory. Similarly, Wang et al. examined the effects of surface roughness, inlet velocity, and other parameters on the VIV characteristics of the riser using a two-way FSI method. The study revealed that both the flow vibration frequency and the transverse vibration frequency of the riser increased significantly with higher inflow velocities and greater surface roughness [9]. The effects of top tension, flow resistance, and VIV responses of the metal and FRP composite risers were examined using finite element simulations [10]. The methods of numerical simulation in Computational Fluid Dynamics offer significant cost-saving advantages in experiments and reduce the experimental workload. By simulating real ocean flow fields through numerical techniques [11], these methods provide a new direction for research. A comparative study of eddy excitation vibrations of three models of metallic and FRP composite marine risers using both 2D and 3D finite element simulation was conducted, which found that the displacement of the FRP composite risers was significantly greater than that of the steel risers, while the stress responses were inversely correlated [12]. The weight-saving advantage of FRP composite risers for ultra-deep ocean application was also proven [13]; however, this reduction in weight might result in considerable deformation of the composite risers, leading to misalignment and instability [14]. A quasi-static analytical model was developed to forecast the fatigue damage of an FRP composite flexible riser under eddy-excited vibrations in a deep-sea environment [15], and a theoretical model based on the energy method was also proposed to predict the fatigue performance of unbonded flexible risers under axisymmetric loading [16]. Kang et al. studied the progressive damage process of composite risers with coupled numerical simulations, and the residual stresses generated during this process were predicted [17]. Edmans et al. proposed a multiscale approach for designing flexible risers with three models, i.e., a nonlinear beam cell, a nonlinear section response model, and a detailed finite element model, to effectively predict the stress values that contribute to fatigue damage in flexible risers [18].

Unlike traditional metal risers, the better performance of FRP composite risers could be achieved by optimizing the structural lay-ups due to the anisotropic properties of FRP composite layers. Wang et al. employed both manual customized design and the surrogate-

assisted evolutionary algorithm (SAEA) method to optimize the structural geometries of FRP composite risers, and the weight savings from the optimization were up to 25% of the orthogonal lay-ups. Both tension and pressure conditions might lead to the failure of FRP composite risers; to address this problem [19], Zheng et al. conducted multi-objective optimization by manipulating the fiber angles and thicknesses of each composite layer to achieve maximum critical pressure and tensile strength [20]. A well-reasoned design of composite risers not only reduced weight but also enhanced performance, making it essential to evaluate their structural integrity [2,3,21]. In order to further investigate the damage mechanisms of composite marine risers in deep-sea environments, experimental studies and numerical analyses were conducted by Pham et al. [22]. It was also proven that the fatigue performance of composite risers under applied forces was significantly influenced by the layup configurations [23]. The frequency-locked resonance phenomenon of composite marine risers was investigated with various composite fiber orientation angles, and it found that the frequency-locked spacing of the risers gradually decreased with the increasing fiber orientations, while the amplitude of vibration correspondingly increased with the increasing fiber orientation angle [24].

More recently, advanced mathematical techniques were employed in research on risers' performance. In a comparative study for predicting the VIV of marine risers with different mathematical techniques, the maximum a posteriori method was proven to be a most effective approach [25]. Similarly, the Grey Theory and multiple linear regression calculation were both employed in the study of the impact of various design parameters on the VIV response of metal and FRP composite risers. [5,22,26] developed an empirical model for prediction of the fatigue life of composite risers by integrating the critical stress values at the fatigue damage locations of the risers. Both scalable and non-scalable modeling methods were employed to study the response of composite risers, and it was found that non-scalable modeling produced spurious high-frequency vibrations under ambient loading conditions [27].

While prior research has examined factors influencing VIV in FRP composite risers, most studies focus on isolated parameters, with scant attention paid to coupled multi-parameter effects [28]. In practical engineering, the VIV of risers emerges from complex nonlinear interactions among multiple variables. Single-factor analyses were proven inadequate for predicting riser dynamics under realistic ocean conditions. To address this gap, a three-factor, three-level orthogonal experimental design coupled with multiple nonlinear regression was implemented to investigate interaction mechanisms among tensile modulus, flexural modulus, top tension, and flow velocity in metallic and FRP risers. The novelty of this work resided in its development of a VIV prediction model that quantified nonlinear coupling relationships through regression analysis. This approach elucidated synergistic parameter interactions, overcame limitations inherent in traditional single-factor analyses, and provided design optimization strategies for marine risers. Beyond its industrial contributions, including advancing understanding of multi-parameter VIV mechanics, this research could also serve as an engineering demonstration case for academic integration into curricula of Engineering Simulation Technology. By transforming industrial challenges into hands-on learning experiences, the study embodies research-driven pedagogy that bridges theoretical knowledge with practical application while cultivating competencies in the application of simulation tools, experimental design, and data analysis.

2. Design and Methodology

2.1. Riser Design

An optimized FRP composite riser, an orthogonal reinforced FRP composite riser, and a metal riser, which satisfied the same design requirements [16,29], were selected

in this study. More specifically, the optimized FRP composite riser (Figure 1) was designed with 2 mm of titanium liner, 3 axial reinforced FRP composite layers (1.7 mm each), 10 alternately $\pm 53^\circ$ reinforced FRP composite layers (1.64 mm each), and 4 hoop reinforced FRP composite layers (1.75 mm each); the orthogonal reinforced FRP composite riser (Figure 2) was designed with 2 mm of titanium liner and 21 alternately axial and hoop reinforced FRP composite layers (1.385 mm each for axial layers and 2.15 mm each for hoop layers). The metal riser's design was relatively simple compared to the composite riser's design, and its inside and outside diameters were 250 mm and 300 mm, respectively.

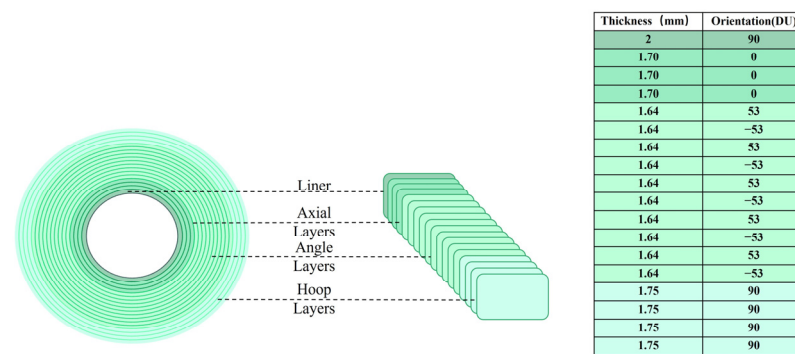


Figure 1. Design configuration of optimized FRP composite riser.

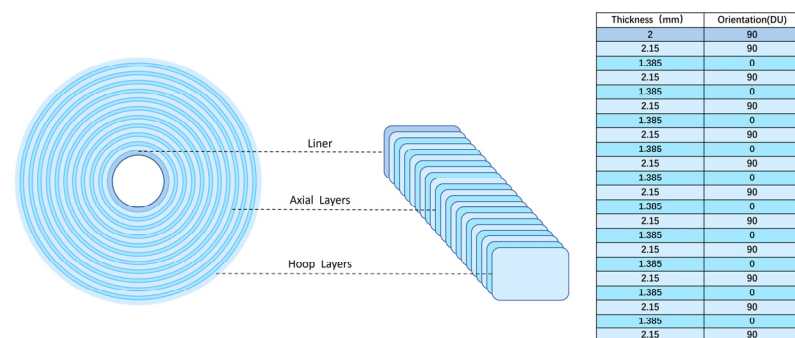


Figure 2. Design configuration of orthogonal FRP composite riser.

The specific parameters of the three risers are shown in Table 1, from which it was found that the unit masses of the optimized FRP composite riser and the orthogonal reinforced FRP composite riser were only 27.0% and 35.3% that of the metal riser. Moreover, the unit mass of the optimized FRP composite riser was about 23.3% lighter than that of the orthogonal reinforced FRP composite riser. Therefore, it is evident that the different designs of FRP composite risers under same design requirement would affect the performance of the riser.

Table 1. Design parameters for optimized FRP composite riser, orthogonal FRP composite riser, and X80 riser.

Type	Number of Layers	Inside Diameter (m)	Outside Diameter (m)	Unit Volume (m ³)	Unit Mass (kg/m)
Optimized FRP composite riser	18	0.25	0.311	0.0269	45.7
Orthogonal FRP composite riser	22	0.25	0.329	0.0359	59.6
X80 riser		0.25	0.3	0.0216	169

2.2. Material Properties

In this paper, AS4 carbon fiber was used as the reinforcing material and epoxy resin was used as the matrix. AS4 carbon fiber has high specific strength, as well as good corrosion, fatigue, and shock resistance, thereby significantly improving the performance and service life of the FRP risers [30–32]. Epoxy resin with excellent bonding properties can be combined well with AS4 carbon fiber to form a strong composite structure and to improve the overall strength and durability of the FRP composite riser [33,34]. In order to achieve the waterproof performance of FRP composite risers, titanium alloy with excellent corrosion resistance and high specific strength was selected as the material for the inner liner. The benchmark metal riser was made of high-grade X80 steel [35,36]. Table 2 presents the specific properties of the used materials in the study.

Table 2. Material properties for FRP composite riser and steel riser.

Material	Density (kg/m ³)	Modulus of Elasticity (GPa)	Shear Modulus (GPa)	Poisson's Ratio	Ultimate Strength (MPa)	Yield Strength (MPa)	Elongation at Break (%)
AS4	1750.00	E1 = 235.00 E2 = E3 = 14.00	G12 = G13 = 28.00 G23 = 5.60	$\nu_{12} = \nu_{13} = 0.20$ $\nu_{23} = 0.25$	3590.00		
AS4–epoxy composite	1530.00	E1 = 135.40 E2 = E3 = 9.37	G12 = G13 = 4.96 G23 = 3.20	$\nu_{12} = \nu_{13} = 0.32$ $\nu_{23} = 0.46$	S1 = 1732, S2 = 49.4, S12 = 71.2		
Epoxy	1200.00	4.50	1.60	0.40	130.00		
Titanium alloy	4430.00	113.80		0.34	950.00	880.00	14.00
Steel-X80	7850.00	207.00		0.30	625.00	555.00	5.87

2.3. Experimental Matrix and Parameters

In addition to the different top tensions and outer diameters of the risers caused by the selected materials and design methods, the length of the riser and the flow velocity were considered, as well. More specifically, 12.5 m, 25 m, and 37.5 m for the riser length were studied and 0.36 m/s, 1.22 m/s, and 2.13 m/s [37,38] were selected as flow velocities.

A full design of a three-factor, three-level experiment would lead to 27 study cases. Considering the large workload and long time period of experiments through the finite element analysis (FEA), an experimental design method of orthogonal tests [39], which would ensure the rationality of the experiment with a relatively small number of tests, was employed and formed 18 study cases for this study. Table 3 shows the study cases with different design parameters.

Table 3. The 18 study cases determined through the orthogonal test.

Case	Design of Experiment			Studied Parameters				
	Riser Type	Riser Length (m)	Flow Velocity (m/s)	Tensile Modulus (GPa)	Flexural Modulus (GPa)	Slenderness Ratio	Velocity (m/s)	Top Tension (N)
1	Optimized FRP riser	12.5	0.36	40.5	35.7	40	0.36	11,200
2	Orthogonal FRP riser	37.5	1.22	50.1	56.8	114	1.22	43,774
3	X80 riser	25	2.13	207	207	83	2.13	62,309
4	Optimized FRP riser	25	1.22	40.5	35.7	80	1.22	22,400
5	Orthogonal FRP riser	12.5	2.13	50.1	56.8	38	2.13	14,591

Table 3. Cont.

Case	Design of Experiment			Studied Parameters				
	Riser Type	Riser Length (m)	Flow Velocity (m/s)	Tensile Modulus (GPa)	Flexural Modulus (GPa)	Slenderness Ratio	Velocity (m/s)	Top Tension (N)
6	X80 riser	37.5	0.36	207	207	125	0.36	93,463
7	Optimized FRP riser	37.5	2.13	40.5	35.7	121	2.13	33,600
8	Orthogonal FRP riser	25	0.36	50.1	56.8	76	0.36	29,183
9	X80 riser	12.5	1.22	207	207	42	1.22	31,154
10	Optimized FRP riser	25	2.13	40.5	35.7	80	2.13	22,400
11	Orthogonal FRP riser	12.5	0.36	50.1	56.8	38	0.36	14,591
12	X80 riser	37.5	1.22	207	207	125	1.22	93,463
13	Optimized FRP riser	37.5	0.36	40.5	35.7	121	0.36	33,600
14	Orthogonal FRP riser	25	1.22	50.1	56.8	76	1.22	29,183
15	X80 riser	12.5	2.13	207	207	42	2.13	31,154
16	Optimized FRP riser	12.5	1.22	40.5	35.7	40	1.22	11,200
17	Orthogonal FRP riser	37.5	2.13	50.1	56.8	114	2.13	43,774
18	X80 riser	25	0.36	207	207	83	0.36	62,309

More specifically, the tensile and flexural modulus were determined using two simple FEA models, as shown in Figure 3. From the displacements obtained with these two models (Table 4), the tensile and flexural modulus were calculated through Equations (1) and (2), respectively.

$$E_{tensile} = \frac{NL}{A\Delta_1} \quad (1)$$

where N is the added axial force, L is the length, A is the riser cross-section area, and Δ_1 is the axial displacement of the model.

$$E_{bending} \approx \frac{PL^3}{3I\Delta_2} \quad (2)$$

where P is the added transverse force, L is the length, I is the moment of inertia $I = \frac{\pi(D_o^4 - D_i^4)}{64}$, and Δ_2 is the transverse displacement of the model.

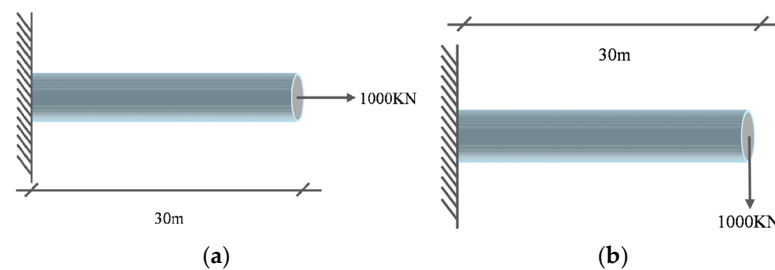


Figure 3. Two elastic modulus calculation models: (a) for tensile modulus calculation, (b) for flexural modulus calculation.

Table 4. Displacement and modulus of elasticity for 18 study cases.

Case	Δ_1 (m)	Δ_2 (m)	E_{tensile} (GPa)	E_{flexural} (GPa)	Case	Δ_1 (m)	Δ_2 (m)	E_{tensile} (GPa)	E_{flexural} (GPa)
1	0.0277	0.9419	40.5	35.7	10	0.0277	0.9419	40.5	35.7
2	0.0141	0.4136	50.1	56.8	11	0.0141	0.4136	50.1	56.8
3	0.0067	0.2112	207	207	12	0.0067	0.2112	207	207
4	0.0277	0.9419	40.5	35.7	13	0.0277	0.9419	40.5	35.7
5	0.0141	0.4136	50.1	56.8	14	0.0141	0.4136	50.1	56.8
6	0.0067	0.2112	207	207	15	0.0067	0.2112	207	207
7	0.0277	0.9419	40.5	35.7	16	0.0277	0.9419	40.5	35.7
8	0.0141	0.4136	50.1	56.8	17	0.0141	0.4136	50.1	56.8
9	0.0067	0.2112	207	207	18	0.0067	0.2112	207	207

The slenderness ratio was defined as the ratio of the length of a riser model to its outer diameter, which was determined based on the riser types (different outer diameters) and the selected riser lengths.

The required top tension force was determined based on the riser types (different densities of materials, different thicknesses of structural layers) and the selected riser lengths (Table 3). Here, we have to note that the required top tension forces of the metal riser and the FRP composite riser were recommended for calculation using 2 times and 1.5 times their structural weights, respectively [25].

Based on the diameter of the riser and the flow rate, the relevant parameters for each condition were determined, as shown in Table 5.

Table 5. Relevant parameters for 18 study cases.

Case	Flow Density (kg/m ³)	ν (m ² /s)	Re	St	I (%)
1	1024	1.06×10^{-6}	105,623	0.20	3.768
2			378,660	0.21	3.213
3			602,830	0.22	3.031
4			357,943	0.21	3.235
5			61,104	0.22	2.996
6			101,887	0.20	3.785
7			624,934	0.22	3.018
8			111,736	0.20	3.742
9			345,283	0.21	3.250
10			624,934	0.22	3.018
11			111,736	0.20	3.742
12			345,283	0.21	3.250
13			105,623	0.20	3.768
14			378,660	0.21	3.213
15			602,830	0.22	3.031
16			357,943	0.21	3.235
17			61,104	0.22	2.996
18			101,887	0.20	3.785

The Reynolds number (Re) is the ratio of inertial and viscous forces in the fluid, which is defined as

$$Re = \frac{\rho UD}{\mu} = \frac{UD}{\nu} \quad (3)$$

where U is the velocity of incoming flow (m/s), D is the diameter of the cylinder (m), and ν is the kinematic viscosity coefficient of a fluid (m²/s).

The Strouhal number (St) is used to describe the effect of periodic oscillations in fluid flow, which is defined as

$$St = \frac{f_s D}{U} \quad (4)$$

where D is diameter of the tube (m), U is the incoming velocity of the flow field (m/s), and f_s is the vortex shedding frequency (Hz).

According to the relationship between Re and St [40], when Re ranges from 300 to 300,000, St is 0.2; when Re ranges from 300,000 to 3,500,000, St is 0.2–0.3.

In the FEA simulation, turbulence intensity (I) is required, and it can be calculated through Equation (5).

$$I = 0.16Re^{(-1/8)} \quad (5)$$

2.4. Numerical Simulation

The numerical simulations of 18 study cases were carried out using Ansys workbench 2022 R1. For FRP composite risers, six modules, i.e., Geometry, ACP (Pre), Transient Structure, Fluid Flow (Fluent), ACP (Post), and System Coupling, were involved (Figure 4a). For the high-grade steel riser, only four modules, i.e., Geometry, Transient Structure, Fluid Flow (Fluent), and System Coupling, were needed.

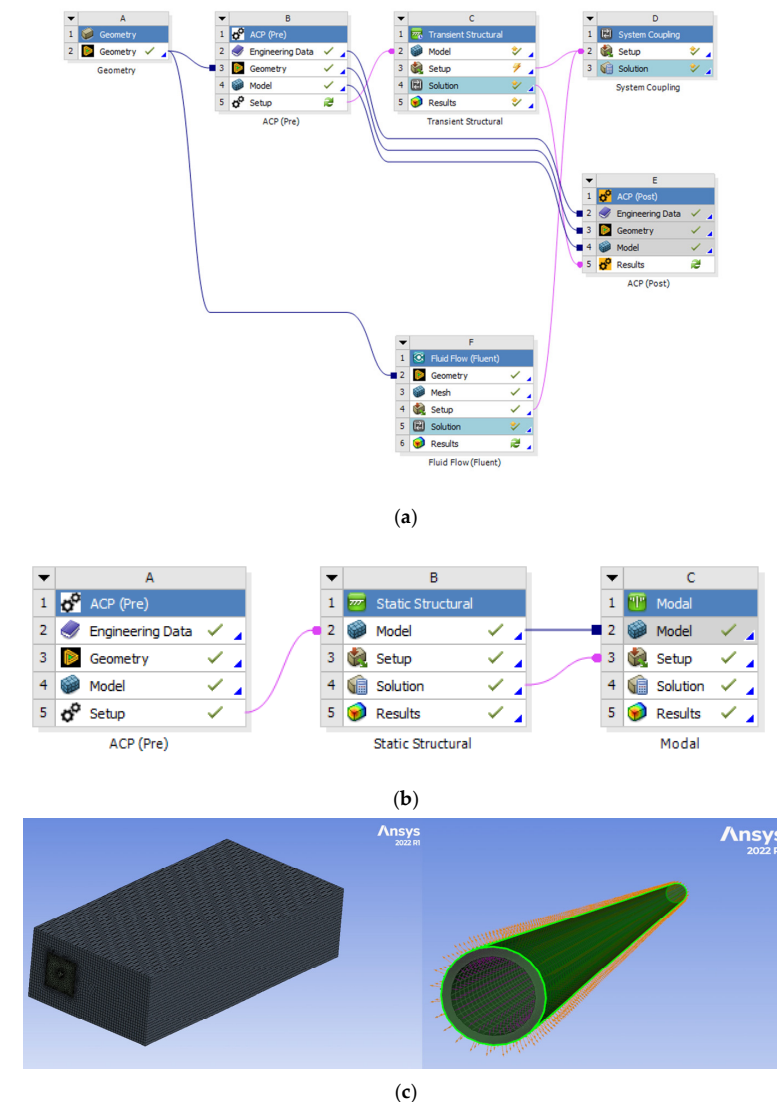


Figure 4. Flow chart and finite element model of the simulation for the FRP composite riser: (a) for the coupled transient analysis, (b) for modal analysis, (c) for the composite riser and the fluid field.

In order to verify the feasibility and accuracy of the FEA model, an experiment using a Polymethyl Methacrylate (PMMA) riser (modulus of 3.5 GPa, length of 1.25 m, inner diameter of 16 mm, and outer diameter of 20 mm) under the conditions of 0.2 m/s flow velocity, a 5 cm wave height, a 1 s wave period, and a fixed–fixed boundary condition [41] (Figure 5) was conducted and compared with a FEA model using this proposed method. From the comparison of strain results at the location 0.6 m from the bottom of the riser model (upstream face), it was observed that $37.71 \mu\epsilon$ was for experiment and $41.69 \mu\epsilon$ was for the proposed FEA simulation, which proved the feasibility and accuracy of the proposed FEA method.



Figure 5. Benchmark test of PMMA riser.

3. Results and Discussion

3.1. Natural Frequencies of Risers for the 18 Different Study Cases

The wet modal analysis of the risers under 18 different cases was conducted using ANSYS software 2022 R1 to determine the natural frequency of the riser in water. For the two composite risers, three modules, ACP (Pre), Static Structural, and Modal, were utilized to obtain the wet natural frequencies (Figure 4b). In contrast, only two modules, Static Structural and Modal, were needed to model the metallic riser for the same purpose. The wet natural frequencies for the 18 cases are presented in Figure 6.

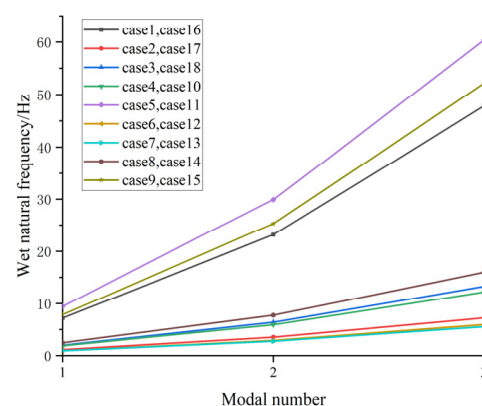


Figure 6. Comparison of wet natural frequencies for 18 study cases.

From Figure 6, it is obvious that the length significantly affected the natural frequency of the riser with same material, i.e., the natural frequency of the riser with 12.5 m was approximately three to four times greater than that with 25 m and seven to eight times greater than that with 37.5 m. When the length of the riser was the same, it was found that the riser's material affected the natural frequency, i.e., the natural frequency of the metal riser was the biggest, followed by those of the orthotropic FRP composite riser and the optimized FRP composite riser.

During the working condition, the frequency of vortex shedding might approach the wet natural frequency of the riser, leading to a phenomenon known as “lock-in”, which would lead to the resonance failure of the riser. In order to avoid this, the reduced velocity (U_r) was calculated to examine the possibility of the riser's lock-in vibration phenomenon using Equation (6). These results are presented in Figure 7, where the natural frequencies from Figure 6 served as the required input parameter.

$$U_r = \frac{U}{f_n D} \quad (6)$$

where U is the flow velocity, f_n is the natural frequency of the riser, and D is the diameter of the riser.

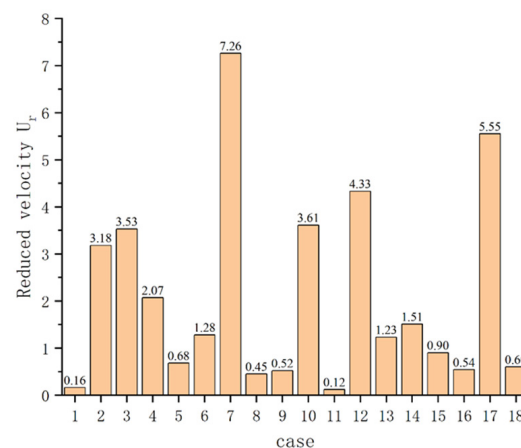


Figure 7. The reduced velocities for the 18 study cases.

Generally speaking, the longer length and bigger flow velocity would increase the possibility of the “lock-in” phenomenon. As illustrated in Figure 7, the reduced velocities for cases 7, 12, and 17 ranged from four to eight, which indicated the possibility of lock-in vibration [42]. However, it has to be noted that the reduced velocity did not include the materials' difference and the effect of fluid–structure interaction.

3.2. Deformations of Risers for the 18 Different Study Cases

Three typical locations in each riser were selected to present the time–history curves of deformations in the downstream direction and the crossflow direction. More specifically, for the riser with 12.5 m, the three typical locations were 2.1 m, 6.3 m, and 10.5 m; for the riser with 25 m, 4.2 m, 12.5 m, and 20.8 m were chosen; for the riser with 37.5 m, they were 6.3 m, 18.8 m, and 31.3 m. In order to present the results more clearly, the deformation data were standardized using the ratio of displacement to the riser's diameter.

3.2.1. Deformations of Risers in the Downstream Direction

The time history of deformations in the downstream direction at each typical location is presented in Figure 8.

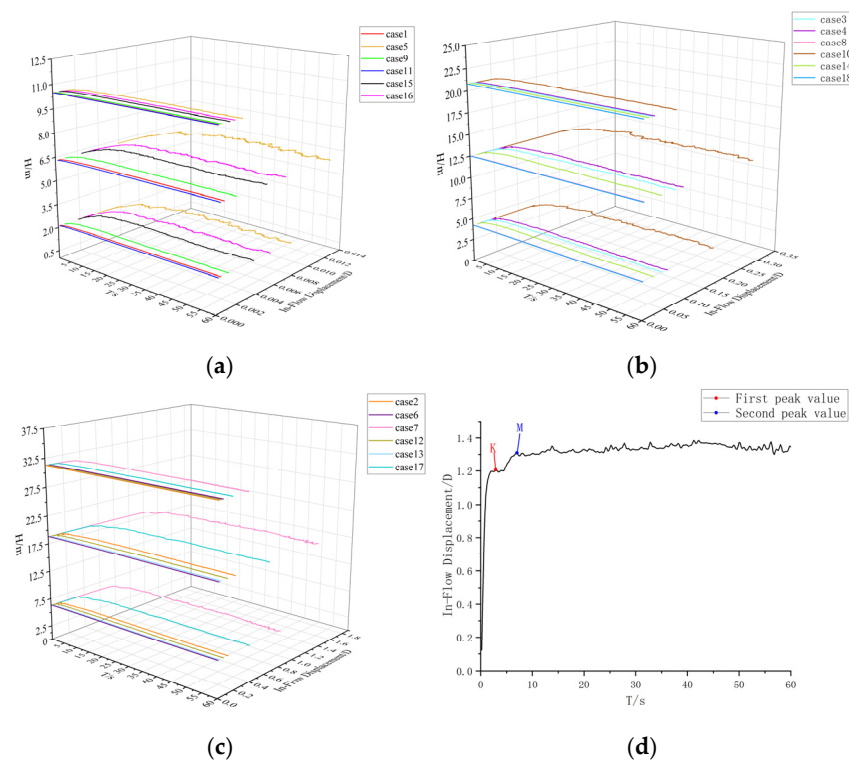


Figure 8. Time history of deformation in the downstream direction: (a) for the riser with 12.5 m, (b) for the riser with 25 m, (c) for the riser with 37.5 m, (d) enlarged example of case 7 at 18.8 m.

Figure 8a illustrates that the maximum standardized deformations were 6.99×10^{-3} (case 16), 9.89×10^{-3} (case 5), and 4.96×10^{-3} (case 15) for the optimized FRP composite riser, the orthogonal FRP composite riser, and the X80 steel riser with a length of 12.5 m. Figure 8b shows that the maximum standardized deformations were 2.81×10^{-1} (case 10), 4.40×10^{-2} (case 14), and 7.79×10^{-2} (case 3) for the optimized FRP composite riser, the orthogonal FRP composite riser, and the X80 steel riser with a length of 25 m. Figure 8c shows that the maximum standardized deformations were 1.385 (case 7), 6.55×10^{-1} (case 17), and 1.18×10^{-1} (case 12) for the optimized FRP composite riser, the orthogonal FRP composite riser, and the X80 steel riser with a length of 37.5 m. We concluded that under the same condition, the downstream deformations of the two types of FRP composite risers were both larger than that of the metal riser, and the optimized FRP composite riser had the largest downstream deformation.

In addition, the maximum downstream deformations for the 18 study cases are illustrated in Figure 9 for the whole length of the risers.

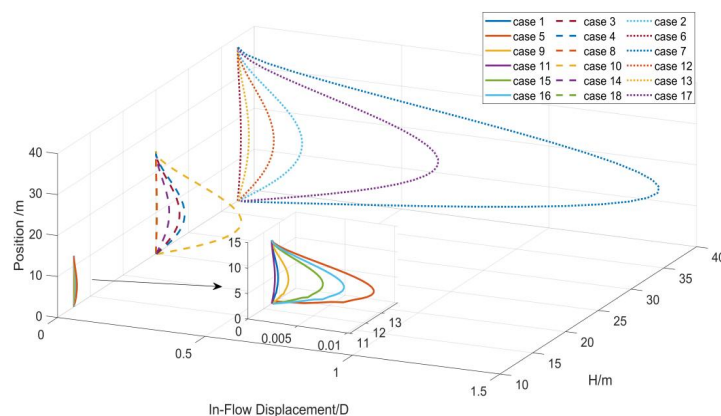


Figure 9. Maximum standardized downstream deformations for all 18 cases (0 m was the top end).

As can be seen in Figure 9, the maximum amplitude downstream occurred in the middle part of the riser, and for the cases with same riser type and flow velocity, the maximum amplitude downstream increased with the increase in riser lengths.

3.2.2. Deformations of Risers in the Crossflow Direction

The time history of deformations in the crossflow direction at each typical location is presented in Figure 10.

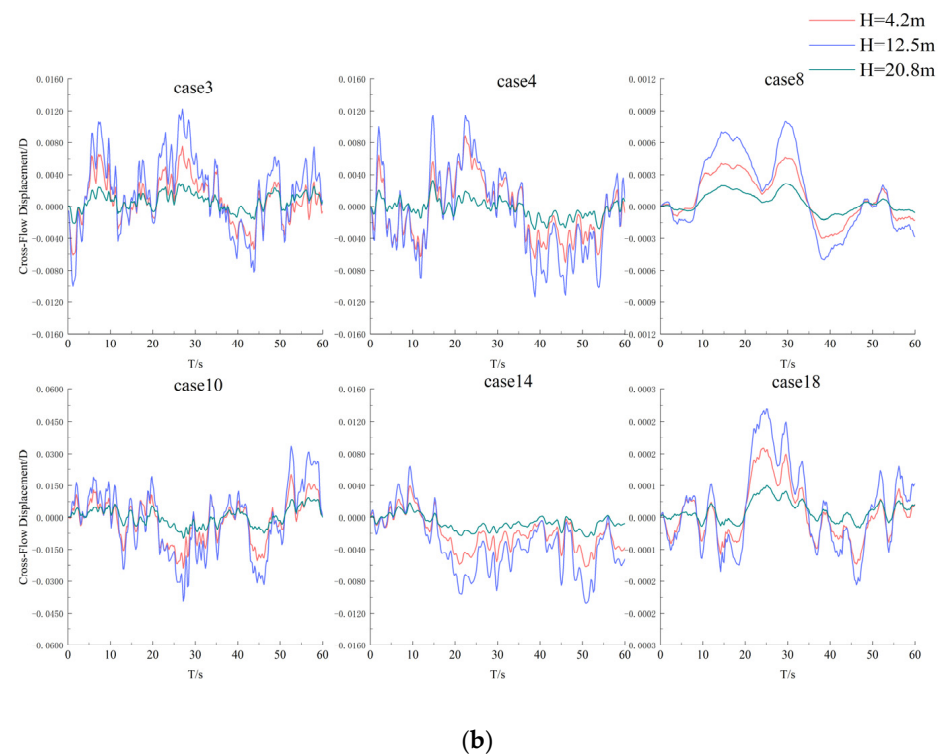
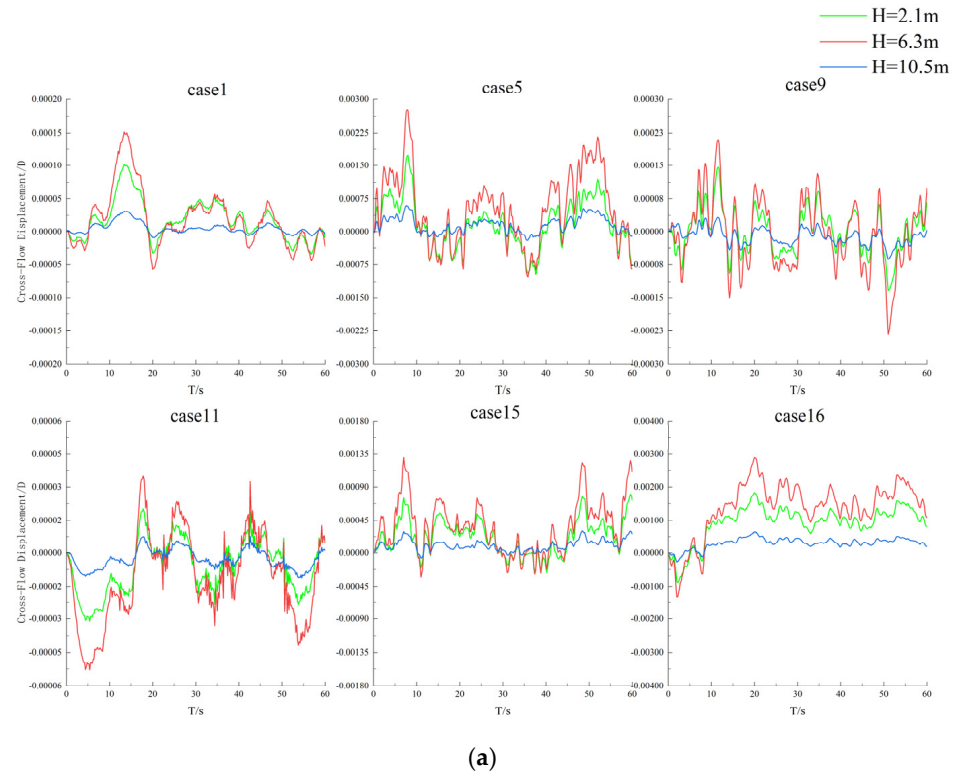


Figure 10. Cont.

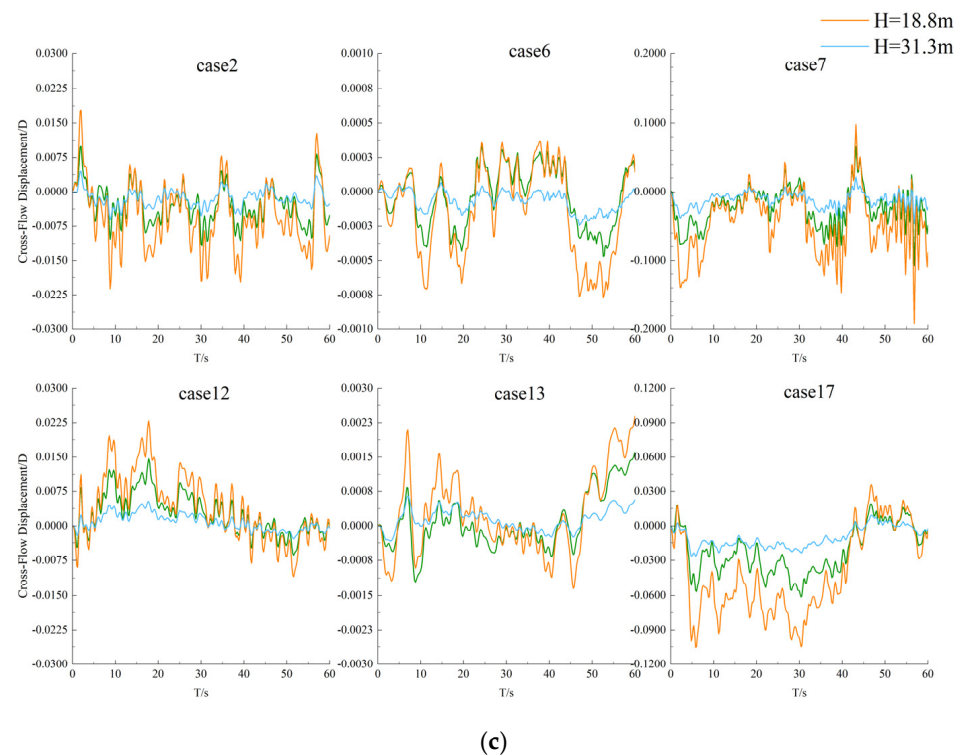


Figure 10. Time history of deformation in the crossflow direction: (a) for the riser with 12.5 m, (b) for the riser with 25 m, (c) for the riser with 37.5 m.

For the three typical locations of the same riser, the middle position of the riser had the largest amplitude and the most obvious vibration in the crossflow direction. Moreover, the vibration in the crossflow direction was more intense than that in the downstream direction. More specifically, for the riser with 12.5 m, the maximum standardized deformations were 2.90×10^{-3} (case 16), 2.77×10^{-3} (case 5), and 1.30×10^{-3} (case 15) for the optimized FRP composite riser, the orthogonal FRP composite riser, and the X80 steel riser (Figure 10a). For the riser with 25 m, the maximum standardized deformations were 4.12×10^{-2} (case 10), 1.12×10^{-2} (case 14), and 1.26×10^{-2} (case 3) for the optimized FRP composite riser, the orthogonal FRP composite riser, and the X80 steel riser (Figure 10b). For the riser with 37.5 m, the maximum standardized deformations were 1.97×10^{-1} (case 7), 1.08×10^{-1} (case 17), and 2.39×10^{-2} (case 12) for the optimized FRP composite riser, the orthogonal FRP composite riser, and the X80 steel riser (Figure 10c).

It can also be concluded that the vibration amplitudes in the crossflow direction of the two types of FRP composite risers were both larger than that of metal riser, and the vibration of the optimized FRP composite riser was the biggest.

In addition, the maximum cross deformations for the 18 cases are illustrated in Figure 11 for the whole length of the risers.

As can be seen in Figure 11, the maximum amplitude in the crossflow direction occurred in the middle part of the riser, and for the cases with same riser type and flow velocity, the maximum amplitude in the crossflow direction increased with the increase in riser lengths.

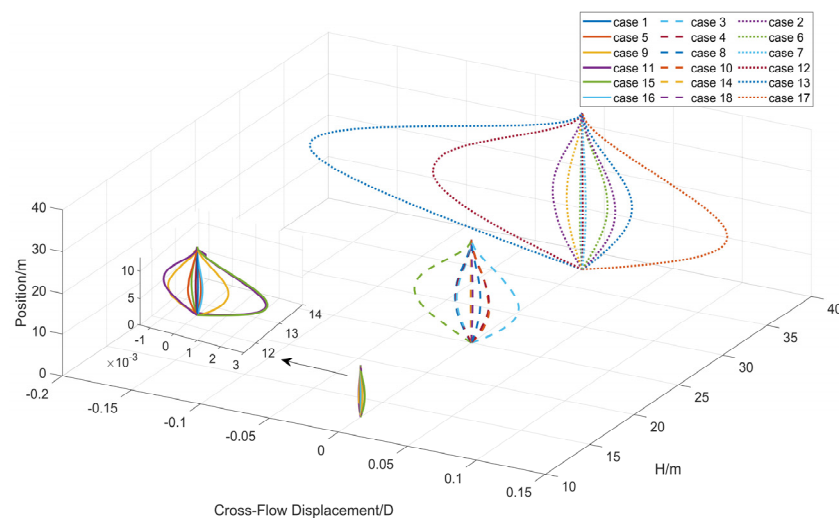


Figure 11. Maximum standardized crossflow deformations for all 18 cases.

3.2.3. Total Deformations of Risers

Based on the results from Sections 3.2.1 and 3.2.2, it was observed that the riser firstly produced a large displacement in the downstream direction and then vibrated in the crossflow direction cyclically. The maximum total deformations and the time duration for all 18 study cases are presented in Figure 12.

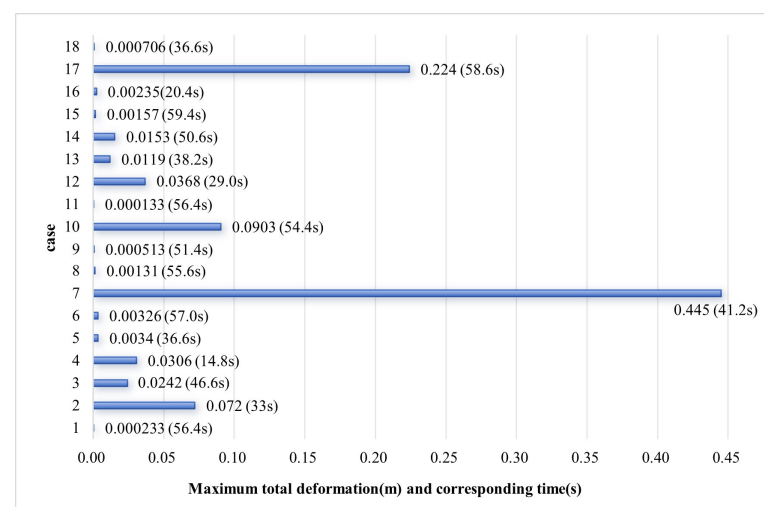


Figure 12. Maximum total deformation and time duration for all 18 study cases.

Based on the various types of marine risers, the 18 study cases were categorized into three groups for the analysis of the results. The maximum total deformation of the orthogonal FRP composite riser occurred in case 17 at 58.6 s, with a result of 0.224 m; the maximum total deformation of the optimized FRP composite riser was found in case 7 at 41.2 s, with a result of 0.445 m; and the maximum total deformation of the X80 steel riser was observed in case 12 at 29.0 s, with a result of 0.0368 m. From these, the importance of the riser's material (modulus) has been proven. Similarly, the maximum total deformations for all three types of risers occurred with the length of 37.5 m, showing the significance of the riser's length.

3.3. Stresses of Risers for the 18 Different Study Cases

The maximum stresses of 25 points in length were selected for each riser to present the stress distribution under the 18 study cases (Figure 13).

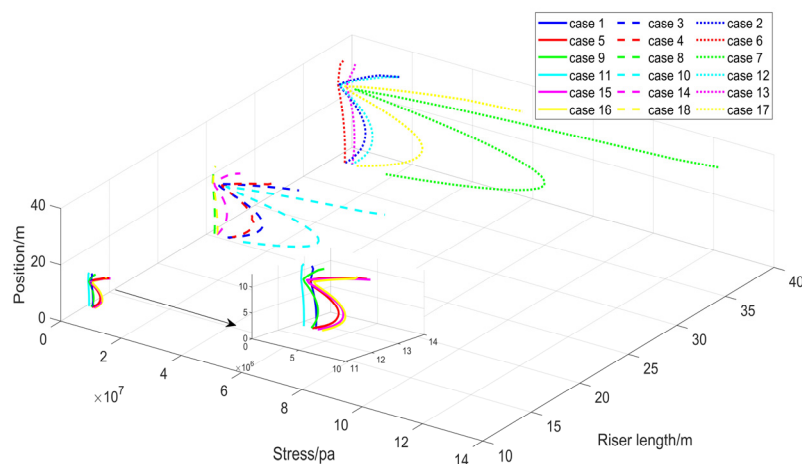


Figure 13. Stress distribution for all 18 study cases (0 m was the top end).

As can be seen from Figure 13, the stress distribution for each type of riser had a similar trend, i.e., the maximum stresses occurred at the bottom end of the riser due to the fixed support, while the minimum stresses occurred at the position of about 1/4 length of the riser from the bottom. Also, it was found that the second maximum stresses appeared at the position of about 1/3–1/2 length of the riser from the top end, where the maximum deformation occurred.

The time history of maximum stresses for 18 cases with different riser lengths is illustrated in Figure 14.

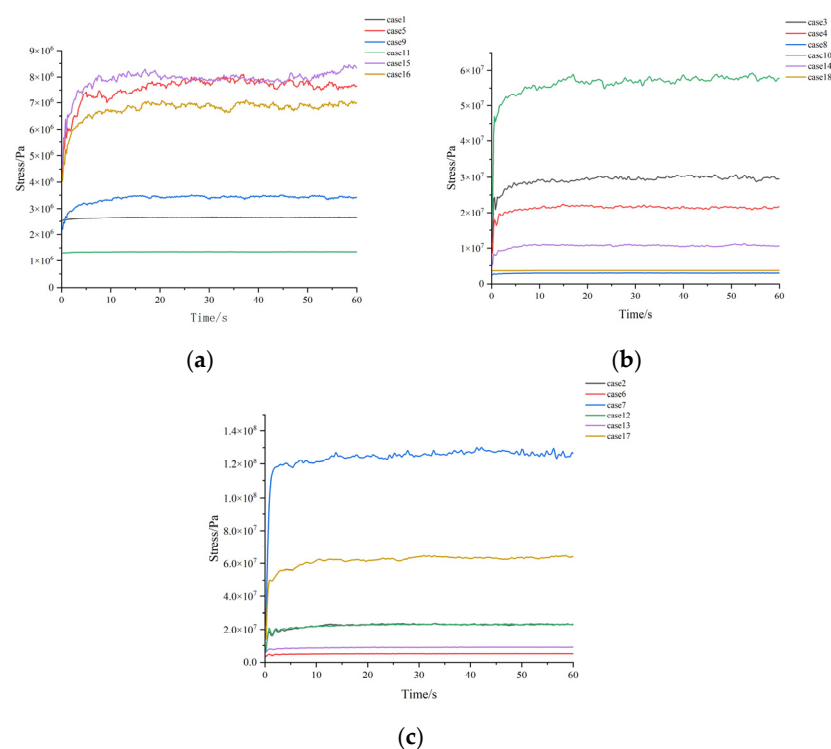


Figure 14. Time history of maximum stresses for 18 study cases: (a) with riser length of 12.5 m, (b) with riser length of 25 m, (c) with riser length of 37.5 m.

From Figure 14, it was found that the maximum stress for each case tended to stabilize after a rapid rise in a short period of time. This trend was similar to that of total deformation, and the time of the occurrence of maximum stress was similar to that of the occurrence of

maximum downstream deformation, which indicated that the downstream deformation was more significant than the crossflow deformation for the riser's stress.

According to the previous study [43], 67% of the yield stress was utilized as the allowance stress (371 MPa) for the X80 riser. For cases 3, 6, 9, 12, 15, and 18, which were X80 risers, the maximum stresses were 29.7 MPa (46.6 s), 6.02 MPa (35 s), 3.36 MPa (26.4 s), 22.2 MPa (33 s), 8.35 MPa (58.4 s), and 4.14 MPa (36.6 s), respectively. Therefore, in all of these cases, X80 risers were safe.

Correspondingly, the global stress distribution could not be used to examine the safety of the FRP composite risers directly. More specifically, the maximum global stresses would be used as the stress condition for the FRP composite risers, and then the allowance stresses (Table 2) would be divided by actual maximum stresses in the fiber direction (S_1), the transverse direction (S_2), and the shear direction (S_{12}) as the safety factor (SF) [44].

For cases 1, 4, 7, 10, 13, and 16, which were optimized FRP composite risers, the minimum SFs in each composite layer were obtained and presented in Figure 15, using case 7 as an example.

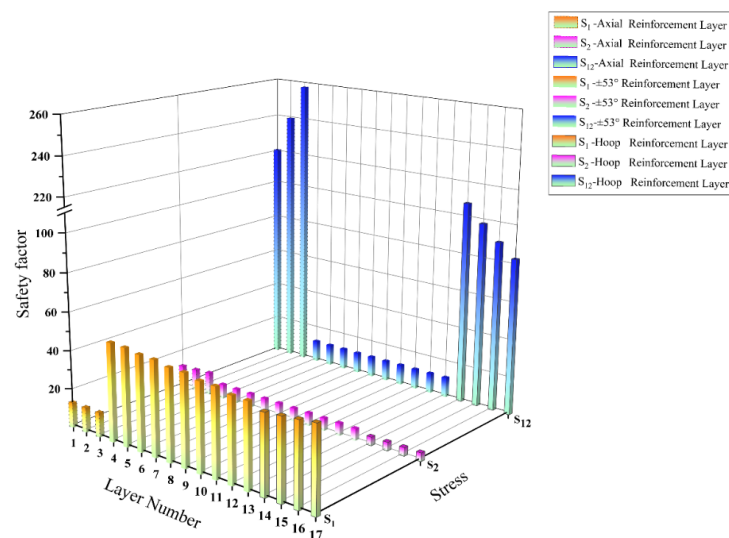


Figure 15. Safety factors in each layer for case 7.

From Figure 15, the minimum SFs for case 7 were 12.85 (layer 3), 4.44 (layer 17), and 10.65 (layer 4) in the fiber direction, the transverse direction, and the shear direction.

In terms of SF in the fiber direction, the axial reinforcement layers had the minimum SFs; the SFs in the $\pm 53^\circ$ reinforcement layers decreased gradually from inside to outside; and the SFs in the hoop reinforcement layers showed an increasing trend from inside to outside. This result indicated that the axial reinforcement layers undertook the main load in the fiber direction.

For the SF in the transverse direction, the hoop reinforcement layers had the minimum SFs; the SFs in the $\pm 53^\circ$ reinforcement layers decreased gradually from inside to outside; and the SFs in the axial reinforcement layers showed an increasing trend from inside to outside. This result indicated that the hoop reinforcement layers undertook the main load in the transverse direction.

It is obvious that the minimum SF in the shear direction occurred in the $\pm 53^\circ$ reinforcement layers, which indicated that the $\pm 53^\circ$ reinforcement layers undertook the main load in the shear direction.

By comparing the SFs in the fiber direction, the transverse direction, and the shear direction, it was found that the SFs in the transverse direction were far less than those in the other two directions. This showed that the stress in the transverse was the main factor

for the failure of each fiber layer of the riser. At the same time, the moderate SFs in the fiber direction proved the effective use of the fiber's strength with the optimized design.

For cases 2, 5, 8, 11, 14, and 17, which were the orthogonal FRP composite riser, the minimum safety factors in each composite layer were obtained and presented in Figure 16, using case 17 as an example.

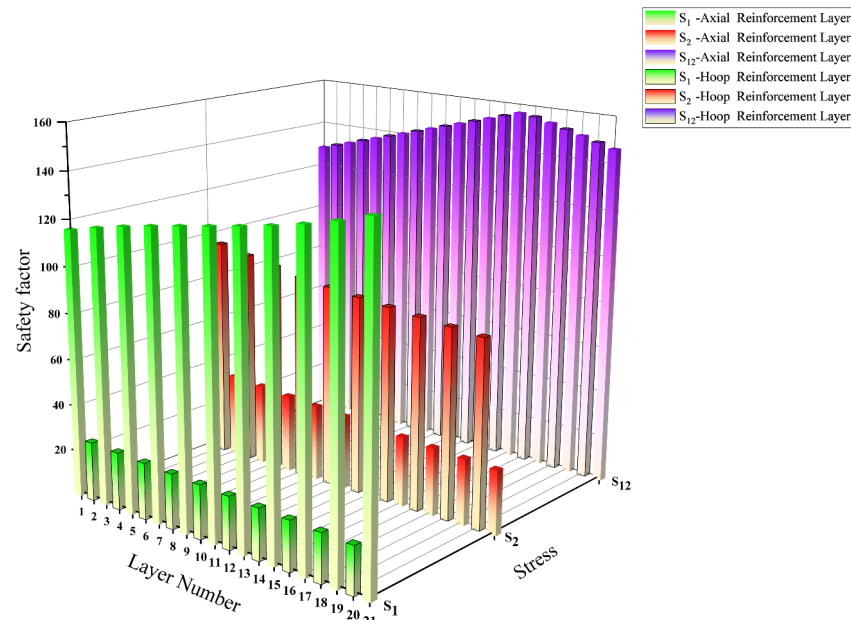


Figure 16. Safety factors in each layer for case 17.

From Figure 16, the minimum SFs for case 17 were 20.59 (layer 10), 28.31 (layer 21), and 128.22 (layer 1) in the fiber direction, the transverse direction, and the shear direction.

In terms of SF in the fiber direction, the axial reinforcement layers presented an increasing trend for the hoop reinforcement layers, while the trend for the axial reinforcement layers was the opposite and the values were much smaller. This result indicated that the axial reinforcement layers undertook the main load in the fiber direction.

For the SF in the transverse direction, the hoop reinforcement layers had the minimum SFs and showed a decreasing trend from inside to outside. This result indicated that the hoop reinforcement layers undertook the main load in the transverse direction.

It is also obvious that the SFs in the shear direction were comparatively large in all of the layers for the orthogonal FRP composite riser.

Based on a comparison of Figures 15 and 16, the smaller SFs (without failure) in the optimized FRP composite riser proved that the optimized design can better utilize the performance of FRP composite materials than the traditional orthogonal design.

3.4. Fitting Analysis with Multiple Nonlinear Regression Model

The VIV of the studied risers involved multiple influencing factors, and the relationship between these factors was nonlinear. Unlike traditional marine risers composed of isotropic materials, FRP composite risers have a layered structure, with anisotropic constituents in each layer. Stress distribution within composite risers varies across individual layers due to reinforcement orientations and layup configurations, and directional stress heterogeneity exists within each fiber layer. Consequently, while equivalent stress is conventionally employed to evaluate the global stress state of FRP composite risers, it is infeasible to directly ascertain stress values within specific fiber layers. In contrast, vibration amplitude constitutes a direct measurable response under VIV and has been consistently utilized as a performance metric in related studies [45–47]. Therefore, the amplitude of

vortex vibration was selected as the target variable in this study. From the analysis of the above sections, it was found that the tensile modulus (E_{tenile}), flexural modulus (E_{flexural}), slenderness ratios (L/D), flow velocities (U), and top tension forces (F) affected the VIV of the studied risers significantly. More specifically, the design methods and materials determined E_{tenile} and E_{flexural} , while the L/D and riser types jointly determined the top tension force.

Therefore, in order to study the coupled effect of these five factors, a fitting analysis with a multiple nonlinear regression model was conducted in this section. The multiple nonlinear regression model is a regression analysis method used to analyze the nonlinear relationship between the independent variables and the target variables. This nonlinear relationship could be modeled using polynomials to fit a predicted function within a reasonable range of the independent variables [48].

3.4.1. Data Handling

The multiple nonlinear regression model is expressed generally in Equation (7).

$$y = f(x_1, x_2, \dots, x_k) + \varepsilon \quad (7)$$

where y is the target variable, $x_1, x_2, x_3 \dots x_k$ is the independent variable, $f(x)$ is a nonlinear function, and ε is the error term.

More specifically, y is the amplitude of vortex vibration in the crossflow direction (m), x_1 is the tensile modulus (GPa), x_2 is the flexural modulus (GPa), x_3 is the slenderness ratios, x_4 is the top tension forces (N), and x_5 is the flow velocities (m/s).

The original data for Equation (7) are presented in Equation (8).

$$y = \begin{pmatrix} 7.01 \times 10^{-5} & 1.31 \times 10^{-2} & \dots & 4.75 \times 10^{-2} & 1.34 \times 10^{-4} \end{pmatrix}$$

$$x = \begin{pmatrix} 40.5 & 50.1 & 207 & \dots & 40.5 & 50.1 & 207 \\ 37.5 & 56.8 & 207 & \dots & 37.5 & 56.8 & 207 \\ 40 & 114 & 83 & \dots & 40 & 114 & 83 \\ 11,200 & 43,774 & 62,309 & \dots & 11,200 & 43,774 & 62,309 \\ 0.36 & 1.22 & 2.13 & \dots & 1.22 & 2.13 & 0.36 \end{pmatrix} \quad (8)$$

In order to make the algorithm's performance more stable and efficient during the model training process, the original data were standardized using Equation (9) and presented as in Equation (10).

$$\begin{cases} y' = \frac{18y(n)}{\sum_{n=1}^{18} y(n)} \\ x'_n = \frac{18x(n)}{\sum_{n=1}^{18} x(n)} \end{cases} \quad (9)$$

$$y' = \begin{pmatrix} 5.86 \times 10^{-3} & 1.10 \times 10^0 & \dots & 3.97 \times 10^0 & 1.12 \times 10^{-2} \end{pmatrix}$$

$$x' = \begin{pmatrix} 0.41 & 0.50 & 2.09 & \dots & 0.41 & 0.50 & 2.09 \\ 0.36 & 0.57 & 2.07 & \dots & 0.36 & 0.57 & 2.07 \\ 0.50 & 1.43 & 1.04 & \dots & 0.50 & 1.43 & 1.04 \\ 0.30 & 1.15 & 1.64 & \dots & 0.30 & 1.15 & 1.64 \\ 0.29 & 0.99 & 1.72 & \dots & 0.99 & 1.72 & 0.29 \end{pmatrix} \quad (10)$$

3.4.2. Modeling

In order to increase the accuracy of the fitting analysis, three functional relationships, i.e., exponential function, logarithmic function, and polynomial function, were utilized, and the accuracy of these functions was compared, as shown in Figure 17.

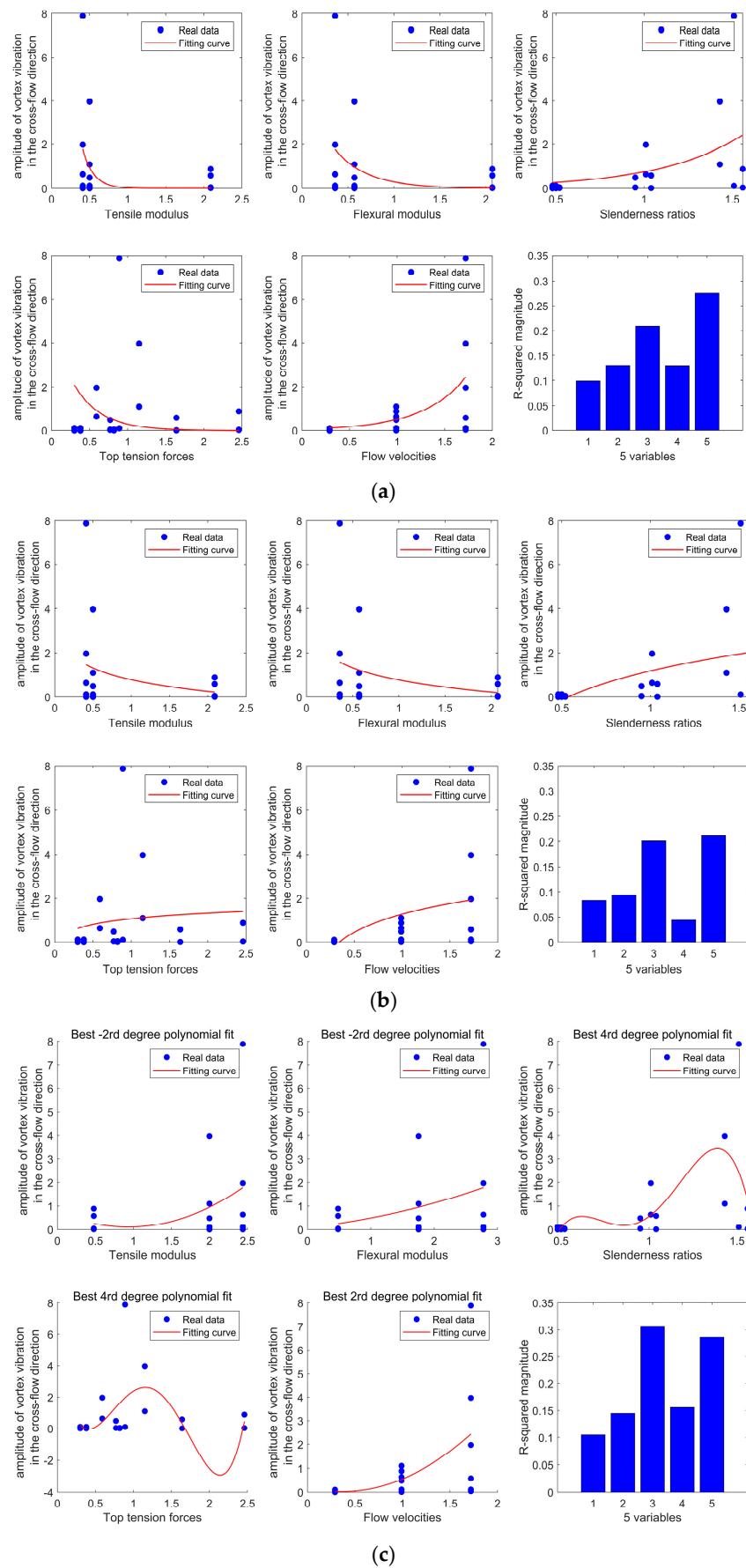


Figure 17. Cont.

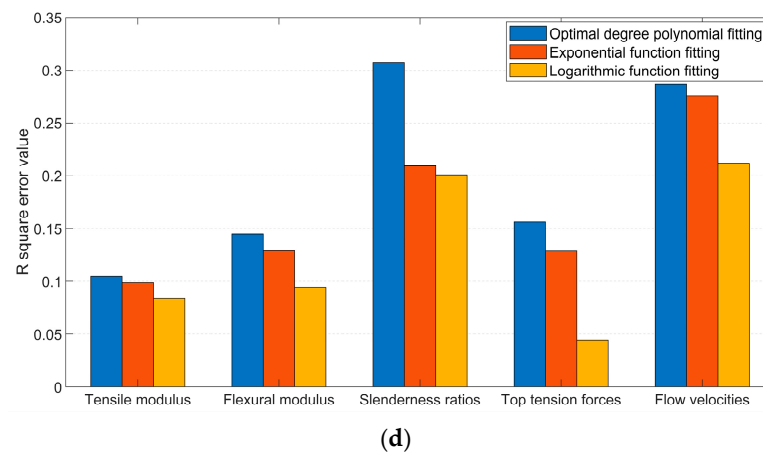


Figure 17. Accuracy of different functional relationships: (a) exponential function, (b) logarithmic function, (c) optimal polynomial function, and (d) R^2 for the three functions.

For polynomial function, the exponential order for each variable should be determined first. Table 6 shows the accuracy of polynomial function with different exponential orders for each variable.

Table 6. The R^2 of polynomial function with different exponential orders for each variable.

	Exponential Order							
	−4	−3	−2	−1	1	2	3	4
R^2 of tensile modulus (x_1)	0.0001	0.0975	0.1186	0.1143	0.0415	0.0556	0.1170	0.0713
R^2 of flexural modulus (x_2)	0.0001	0.0975	0.1386	0.1259	0.0415	0.0556	0.1170	0.1343
R^2 of slenderness ratio (x_3)	0.2183	0.2234	0.2012	0.2789	0.1771	0.1556	0.1547	0.3108
R^2 of top tension force (x_4)	0.0836	0.1032	0.1142	0.1047	0.1033	0.1040	0.1047	0.1683
R^2 of flow velocity (x_5)	0.0779	0.1047	0.1047	0.1047	0.0915	0.2893	0.1047	0.1047

From Table 6, the optimal exponential orders for x_1 , x_2 , x_3 , x_4 , and x_5 were −2, −2, 4, 4, and 2, respectively.

As can be seen from Figure 17, the largest R^2 for each variable was obtained using polynomial function, which will be used in the following part.

3.4.3. Fitting Method and Process

The data were divided into a training database and a test database, i.e., the first 16 sets were the training sets and the last 2 sets were used for the test. Polynomial function fitting used the nonlinear least square method with the criterion of minimizing the sum of squares of the errors [49], as seen in Equation (11).

$$\sum_{i=0}^n (y_i - f_i(x))^2 \quad (11)$$

where y_i was the i -th actual value and $f_i(x)$ was the i -th predicted value.

For the calculation of the nonlinear least square method in this paper, the Levenberg–Marquardt (LM) method, for which the core idea is to solve the nonlinear minimization problem by introducing a damping factor [50], was chosen.

Based on the optimal exponential orders for the five intendent variables from Table 6, the multivariate nonlinear model equation was established as Equation (12). And, using Equation (11), the error equation can be expressed as Equation (13).

$$Y_1(x) = b_1x_1^{-2} + b_2x_2^{-2} + b_3x_3^4 + b_4x_4^4 + b_5x_5^2 + b_0 \quad (12)$$

$$r(x) = \sum_{i=1}^{18} [y_i - Y_1(x)]^2 \quad (13)$$

The initial parameter vectors ($b_0, b_1, \dots, b_5$) were all set to 1 in the first iteration. After 62 iterations, the $r(x)$ remained stable at about 1.89 (Figure 18), and the values for $b_0, b_1, \dots, b_5$ are presented in Table 7. It corresponded to $R^2 = 0.604$ and $RSEM = 1.404$.

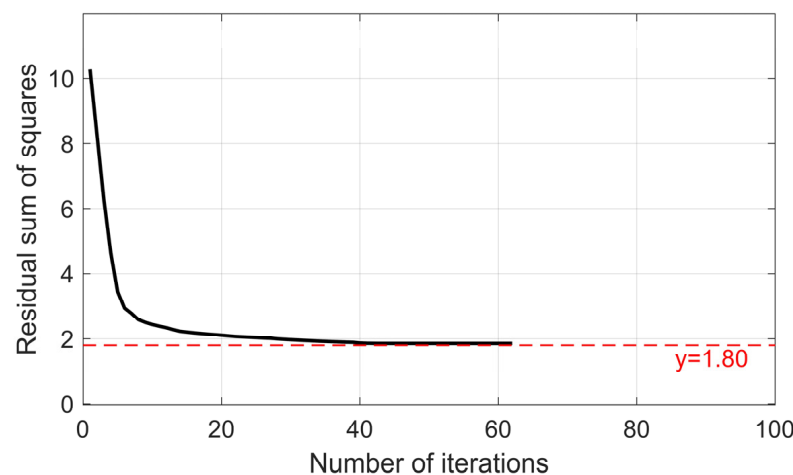


Figure 18. The values of the error equation with the increasing number of iterations of the basic fitting model.

Table 7. The coefficient values of iterations of the basic fitting model.

Coefficient	b_0	b_1	b_2	b_3	b_4	b_5
Fitted values	−0.225	1.3089	−3.091	0.622	0.826	0.686

From the values of R^2 and $RSEM$, it could be concluded that a single factor could not have a significant independent impact on the vibration amplitude in the crossflow direction. Therefore, the coupled effects of two factors have been developed, and the initial Equation (12) has been expanded as Equation (14).

$$Y_2(x) = b_1x_1^{-2} + b_2x_2^{-2} + b_3x_3^4 + b_4x_4^4 + b_5x_5^2 + \sum_{i=1}^4 b_{i+5}x_ix_{i+1} + \sum_{i=1}^3 b_{i+9}x_ix_{i+2} + \sum_{i=1}^2 b_{i+12}x_ix_{i+3} + b_{15}x_1x_5 + b_0 \quad (14)$$

where b_n ($n = 1, 2, \dots, 15$) were the parameters to be optimized.

The values of the error equation (Equation (13)) with the increasing number of iterations are presented in Figure 19.

From Figure 19, it was found that after 83 iterations, the $r(x)$ remained stable at about 0.64, and the values for $b_0, b_1, \dots, b_{15}$ are presented in Table 8. It corresponded to $R^2 = 0.989$ and $RSEM = 0.039$, which indicated the accuracy of the fitted model, and the comparison between the actual vibration amplitude and predicted vibration amplitude is illustrated in Figure 20.

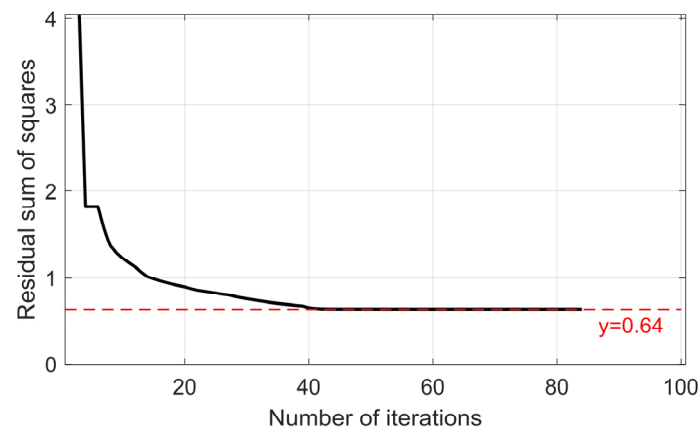


Figure 19. The values of the error equation with the increasing number of iterations of the high-order interaction polynomial model.

Table 8. The coefficient values of the high-order interaction polynomial model.

Coefficient	b_0	b_1	b_2	b_3	b_4	b_5	b_6	b_7
Fitted values	1.527	−7.662	3.985	6.202	3.3473	3.669	−0.893	8.559
Coefficient	b_8	b_9	b_{10}	b_{11}	b_{12}	b_{13}	b_{14}	b_{15}
Fitted values	9.322	−2.110	10.375	16.487	−4.230	−5.069	4.382	−1.655

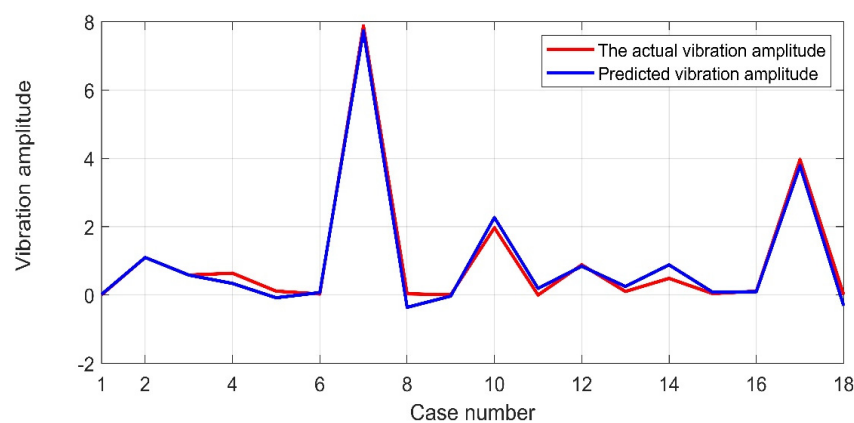


Figure 20. The comparison between the actual vibration amplitude and the predicted vibration amplitude.

From the comparison between the actual vibration amplitude and the predicted vibration amplitude in Figure 20, the overall difference between actual values and predicted values was 4.5%, especially for the cases with obvious VIV. This showed the good accuracy of Equation (14), especially for the cases with obvious VIV. The analysis of the coupled effect of three, four, and five factors was also conducted; however, the R^2 and RSEM did not change obviously, which meant that consideration of the coupled effect of two factors reached an acceptable prediction. It should be noted that based on the technical foundations of nonlinear regression modeling, the method demonstrates robust predictive performance for continuous data within the calibrated parameter range [51]. However, it also exhibits significant limitations when extrapolating to parameter boundary regions, particularly manifesting as substantial prediction errors at the lower critical boundary [51]. Due to this constraint, additional fitting procedures are normally required for design parameters extending beyond the training domain [52]. Future investigations could consider advanced methodologies, such as artificial neural network (ANN). It exhibits enhanced nonlinear modeling capability and generalization ability that effectively capture complex

fluid–structure interactions and efficiently process high-dimensional data, as validated in a recent riser fatigue damage prediction study [53].

Based on the values in Table 8, it was found that the most important single factors were b_1 (tensile modulus) and b_3 (slenderness ratio) and the most significant interacting factors were b_{11} (flexural modulus + top tension force). And, it has to be noted that the interaction of coupled factors generally affected the crossflow VIV more remarkably than a single factor. Within engineering practice, this model could be integrated throughout the design–evaluation cycle. During conceptual design, engineers input core parameters (e.g., environmental conditions, material properties) to delineate viable operating envelopes meeting VIV performance requirements. In the detailed design stage, the model’s output—employed in tandem with finite element analysis—guides parametric optimization while minimizing physical testing needs. For safety assessment, the model predicts riser dynamics across diverse marine environments, establishes data-driven early-warning thresholds, and enables evidence-based decision making across all project phases.

4. Conclusions

In this paper, the vortex-induced vibration characteristics of FRP composite risers (both orthotropic and optimized design) and metal risers under 18 different study cases were investigated using FEA simulations, and the feasibility and accuracy of the FEA method have been proven through a corresponding experimental test. The influence of five factors on the VIV of risers was obtained using a multivariate nonlinear fitting model, and the coupling effect of parameters with different numbers of interactions was also examined.

From this study, the following was found.

- (1) All three types of risers tended to stabilize after the initial growth of displacement in the downstream direction, while the displacement exhibited obvious vibration characteristics in the crossflow direction. With the same condition, the optimized FRP composite riser had the largest displacement, while that of the metal riser was the smallest. Moreover, the increase in the riser’s length and flow velocity, as well as the decrease in the riser’s modulus, led to the increase of downstream displacement and crossflow vibration amplitude.
- (2) The overall stresses of both FRP composite risers were less than that of the metal riser, and the optimized FRP composite riser could more effectively use the strength of the reinforced fiber.
- (3) When the number of coupled parameters reached two, the accuracy of the established multivariate nonlinear fitting model gained an acceptable level. The most important single factors were b_1 (tensile modulus) and b_3 (slenderness ratio), and the most significant interacting factors were b_{11} (flexural modulus + top tension force). Moreover, the crossflow VIV was affected by the interaction of coupled factors, which was generally more remarkable than that of the single factors.

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